# Finding the optimal allocation to maximize the Beta of a portfolio to respect Value at Risk arbitrary constraint

A Preprint

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June, 2026

## Abstract

Our base intuition is to maximize the aggressiveness of a given portfolio, represented by the $\beta$ while minimizing tail risks, that is to say the Value at Risk. The idea is to benefit as much as possible from market momentum and be able to survive a temporary reversal without abandoning the position. We could somehow say that we want an asymmetric $\beta$ that helps to grow during uptrend, but that does not imply high losses during crisis period. Assuming that, on the long-term, markets are increasing, this strategy aims at benefiting from that long-term uptrend and not be taken out during crisis period. Actually, the objective is to determine the optimal allocation that satisfies this strategy.

We assume that asset picking is already done successfully. We scrap five years of daily market data of each of the chosen assets, we define the given parameters and constraints. In our example, the assets are: Tesla, Airbus, Google, Apple, Air France, Ferrari, BMW, LVMH, Société Générale, Goldman Sachs, JPMorgan, Microsoft, General Dynamics, and Boeing. We let the portfolio value be \$500 and we allow for a maximal loss of \$50 under a confidence level $\alpha$ of 99%. We compute risk measures through a KDE (with a Gaussian kernel) with Monte Carlo method. Then we use an SLSQP algorithm that solves our objective under the constraints given. ... Empirically, we find that...

***Keywords*** Portfolio optimization · Kernel Density Estimator · Beta optimization · Value at Risk

## 1 Introduction

First of all, we shall define clearly the steps of our global optimization strategy. We should mention that our work is not about the optimization process itself but rather about the optimization objective and constraints.

The hypothesis we are formulating is that on the long-term, markets are rising. However, investing on the long-term opens the risk of being exposed to crisis. As we have seen throughout the years, and even in recent history, financial markets are exposed to major crisis. Hence, the objective is to take advantage of global markets growth and at the same time being able to survive major crisis.
In finance, we usually define risk as volatility. But it doesn't give any information about black swan events. Value at Risk and Expected Shortfall give a real idea of these events. This is why the constraints will be mostly about minimizing these risk measures as the constructed-portfolio primary objective is to survive crisis. Moreover, this portfolio is also constructed to be aggressive (i.e. $\beta > 1$) so that it profits from growth even more that the benchmark.

Finally, our major contribution is to find the right balance between taking advantage of markets' uptrend, and minimizing black swan events' risk.

Before going further, let's give some definitions. Value at Risk can be defined as: "a measure of the worst expected loss on a portfolio of instruments resulting from market movements over a given time horizon and a pre-defined confidence level". Formally, it can be written as:

$$VaR_\alpha = \inf\{l \in \mathbb{R} : \mathbb{P}(L \geq l) \geq l - \alpha\}$$

Expected Shortfall (also called Conditional Value at Risk—CVaR—) is defined as: "”a measure of the average of all potential losses exceeding the VaR at a given confidence level". Mathematically, we can write it as:

$$CVaR_\alpha = \mathbb{E}[L|L \geq VaR_\alpha]$$

As a reminder, under Basel III regulation, the confidence level for VaR is $\alpha$ = 99%. Through the FRTB, CVaR is observed at 97.5% level. This shift from VaR(99%) to CVaR(97.5%) is a way to strengthen the capital requirements as empirically we always find that CVaR(97.5%) is greater than VaR(99%).

To ensure a minimum diversification, we impose two allocation constraints: a minimum weight for each asset, and a maximum weight as well. The main constraint is about the maximum acceptable Value at Risk (MAVaR).

## 2 Kernel Density Estimation (KDE) in Return Distributions

Using a previous work we established, to measure these risk measures we will use a Kernel Density Estimator (KDE) with Monte Carlo method. The formal mathematical formula for a KDE is:

$$\hat{f}(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{1}$$

Where $h$ is the bandwidth, $K(.)$ the Kernel Function. From our previous study, we find that empirically, a Gaussian kernel is pretty accurate for our purposes. Gaussian kernel is written as follow:

$$K(u) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}u^2} \tag{2}$$

Hence we have:

$$\hat{f}(x) = \frac{1}{nh\sqrt{2\pi}}\sum_{i=1}^{n} e^{-\frac{1}{2}\left(\frac{x-x_i}{h}\right)^2} \tag{3}$$

To decide the value of the bandwidth, we use the Silverman’s Rule of Thumb that imposes:

$$h = \left(\frac{4\hat{\sigma}^5}{3n}\right)^{\frac{1}{5}}$$

The benefit of this method is that it is non-parametric and then it doesn’t impose any arbitrary choice on parameters, it fits itself around the distribution curve.

With an empirical backtesting [1], we observe that among the tested approaches KDE is the one with the lowest violation rate. It has higher values for each measures and we can debate wether it is too conservative or not, but we assume here that is it sufficient for our study. The main importance is to avoid the underestimation of tail risks that a classic gaussian approach implies.

## 3 Market Exposure and Empirical Beta Calibration

### 3.1 Benchmark selection and data acquisition

As US markets are the most considered ones and that S&P500, encompassing most of the biggest companies, provides a broad, highly liquid and widely accepted baseline for capturing systematic risk (market exposure) across our chosen equities. For that reasons, it will be used as our benchmark. The data are acquired on an observation window from June 2021 to June 2026, providing five years of daily observations (roughly 1,250 observations). Data acquisition is done through Python with the "yfinance" library. Values retained are the closing prices.

### 3.2 Covariance-based beta estimation

$\beta$ quantifies an individual asset’s sensitivity to non-diversifiable market movements. To make it simple, a portfolio with $\beta = 1$, will move exactly the same as the market. A portfolio with $\beta = 2$ will have the same variations as the benchmark but two times higher. If $\beta \geq 1$, we say that the portfolio is aggressive, whereas if $\beta \leq 1$, we say that the portfolio is defensive.

Mathematically, it is calculated as the ratio of the covariance between the asset's return and the benchmark's return to the variance of the benchmark's return.

$$\beta_i = \frac{Cov(R_i, R_m)}{Var(R_m)} \tag{4}$$

This measure is done over realized (historical) data. In our study, $\beta_i$ are estimated over the last five years. The formula is relatively easy to find (See Appendix X.) and is obtained through the Ordinary Least Squares econometric method. By computing a single and static beta across the multi-year window, the model assumes that the covariance structure and the linear relationship between the equities and the market are relatively time-invariant. These empirical $\beta_i$ values are extracted to serve as the constant coefficients for the linear objective function in your optimization engine.

# 4 Optimization framework

## 4.1 Objective function

## 4.2 Linear and boundary constraints

## 4.3 Cross-sectional aggregation mechanics

# 5 Non-linear risk constraint (methodological contribution)

## 5.1 Stochastic convergence problem

## 5.2 Freezing the state space (deterministic simulation)

## 5.3 Tail Risk extraction

# 6 Empirical results and performance analysis

## 6.1 Optimal allocation profile

## 6.2 Marginal Risk and Beta Contributions

## 6.3 Distributional analysis

## 6.4 Historical trajectory

# 7 Conclusion

# References

[1] Mazaud. Var_cvar_model: Var_cvar_simulation.py. `https://github.com/GianniMzd/VaR_CVaR_model/blob/main/VaR_CVaR_simulation.py`, 2026.